



United International University
Department of Computer Science and Engineering
Final Examination, Spring 2025
EEE 4261: GREEN COMPUTING

Total Marks: 40

Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. Right margin indicates the marks.

1. [CO1] A tech company releases a new line of laptops branded as "EcoSmart Devices." In its marketing materials, the company claims these laptops are "Environmentally Friendly" and have "Zero Environmental Impact." The promotional content highlights a sleek design with minimal packaging and uses green-themed visuals to reinforce the eco-conscious image. However, a recent investigative report reveals that the laptops are built with non-recyclable materials, the manufacturing process consumes high levels of energy from non-renewable sources, and there is no environmental certification or data to support their claims. [5]

What key issue or practice is being illustrated in this scenario? **Identify** the misleading elements in the company's campaign, and discuss the potential consequences of this type of marketing practice for consumers, the environment, and the company itself.
2. [CO2] A mid-sized software company, *CodeNova*, recently decided to upgrade its infrastructure to become more energy-efficient. The company replaced its old servers with energy-efficient cloud-based solutions, switched all office lighting to LED systems, and programmed all devices to enter sleep mode after periods of inactivity. Additionally, they introduced an employee awareness program on energy-saving habits. Six months after implementation, *CodeNova* reported a 25% reduction in energy consumption and a noticeable drop in their electricity bills. [5]

Discuss how *CodeNova's* implementation of energy-efficient practices positively impacts both the environment and the company. In your answer, explain the environmental benefits of reduced energy consumption and how such initiatives contribute to sustainable development.
3. [CO1] The government of a developing country, *Zarovia*, after reviewing the latest IPCC reports, initiates a national policy aiming to align with the global effort to keep temperature rise below 2°C—ideally targeting 1.5°C. The policy includes transitioning to renewable energy, improving public transportation, enforcing stricter emissions standards for industries, and investing in energy efficiency for urban infrastructure. While some sectors welcome the move, others—particularly in the fossil fuel industry—argue that the transition could slow economic growth and increase unemployment in the short term. [5]

Discuss the significance of the 2°C warming limit in the context of *Zarovia's* climate policy. Why is staying below this threshold critical for both global and national sustainability?

4. [CO2] ABC Tech has recently announced its commitment to becoming carbon-neutral. The company decided to shift to **green devices** and work only with **green-certified suppliers**. It begins designing laptops and smartphones using recyclable materials, energy-efficient components, and modular parts to extend product lifespan. ABC Tech also revises its supply chain policy to only partner with suppliers who meet strict environmental standards—such as using renewable energy, minimizing water waste, and maintaining ethical labor practices. While the move increases production costs slightly, it enhances the company's brand image and meets growing consumer demand for environmentally responsible products. [5]
- Analyze** how this focus on green devices and green suppliers contributes to environmental sustainability. What are the potential long-term benefits and challenges of this approach from both an environmental and business perspective?
5. [CO3] Consider a multinational electronics corporation, "*GlobalTech Inc.*", which previously based its corporate social responsibility program on the Millennium Development Goals (MDGs). With the global transition to the more comprehensive Sustainable Development Goals (SDGs), the company faces increasing pressure to address issues like e-waste from its products and labor conditions in its supply chain. In this context, evolve the company's outdated approach by **proposing** a robust green computing strategy that addresses the following SDGs. i.) **SDG-12**: Responsible Consumption and Production ii.) **SDG-8**: Decent Work and Economic Growth [5]
6. [CO2] *AetherCloud Solutions* is a rapidly expanding cloud service provider. They are planning a new flagship data center at Notun Bazar to manage its growth. The company's key objectives are to significantly reduce the high operational costs of its current facilities and to meet the growing demand from corporate clients for environmentally sustainable services. [5]
- As the lead Green Computing consultant for this project, **provide** a proposal for designing this green data center, focusing your recommendations on the following two foundational areas: **Building infrastructure and Power Supply Considerations**.
7. [CO1] What is Embodied Energy? **Discuss** how understanding and reducing a product's embodied energy can be a key strategy for lowering its overall greenhouse gas emissions. [5]
8. [CO3] *Sajol Agro & Farms*, a large-scale agricultural enterprise, is implementing smart farming practices to improve sustainability by reducing water and fertilizer usage. To do this, they have deployed a vast network of IoT sensors across their fields to collect real-time data on soil moisture, nutrient levels, and weather conditions. However, continuously streaming this massive volume of data to a distant central cloud for analysis is causing high network latency, delaying critical actions like precision irrigation, and resulting in significant energy consumption and data transmission costs. [3+2]
- a) As a Green Computing consultant, **propose** a solution to address these latency, cost, and energy consumption issues.
- b) **Explain** how your proposed architecture differs from a traditional cloud-centric model. What specific benefits does your solution offer in terms of energy efficiency and resource optimization for Verdant Valley Farms?